2ème partie

Méthodes de calcul à l'échelle atomique

Applications potentielles de la théorie de la fonctionnelle de la densité

version 12/04/2026

context = physical chemistry and atomic-level description of molecules and solids

uses computer simulations to assist in solving chemical problems
(methods of theoretical chemistry, incorporated into efficient computer programs, to calculate the structures and properties of molecules and solids)

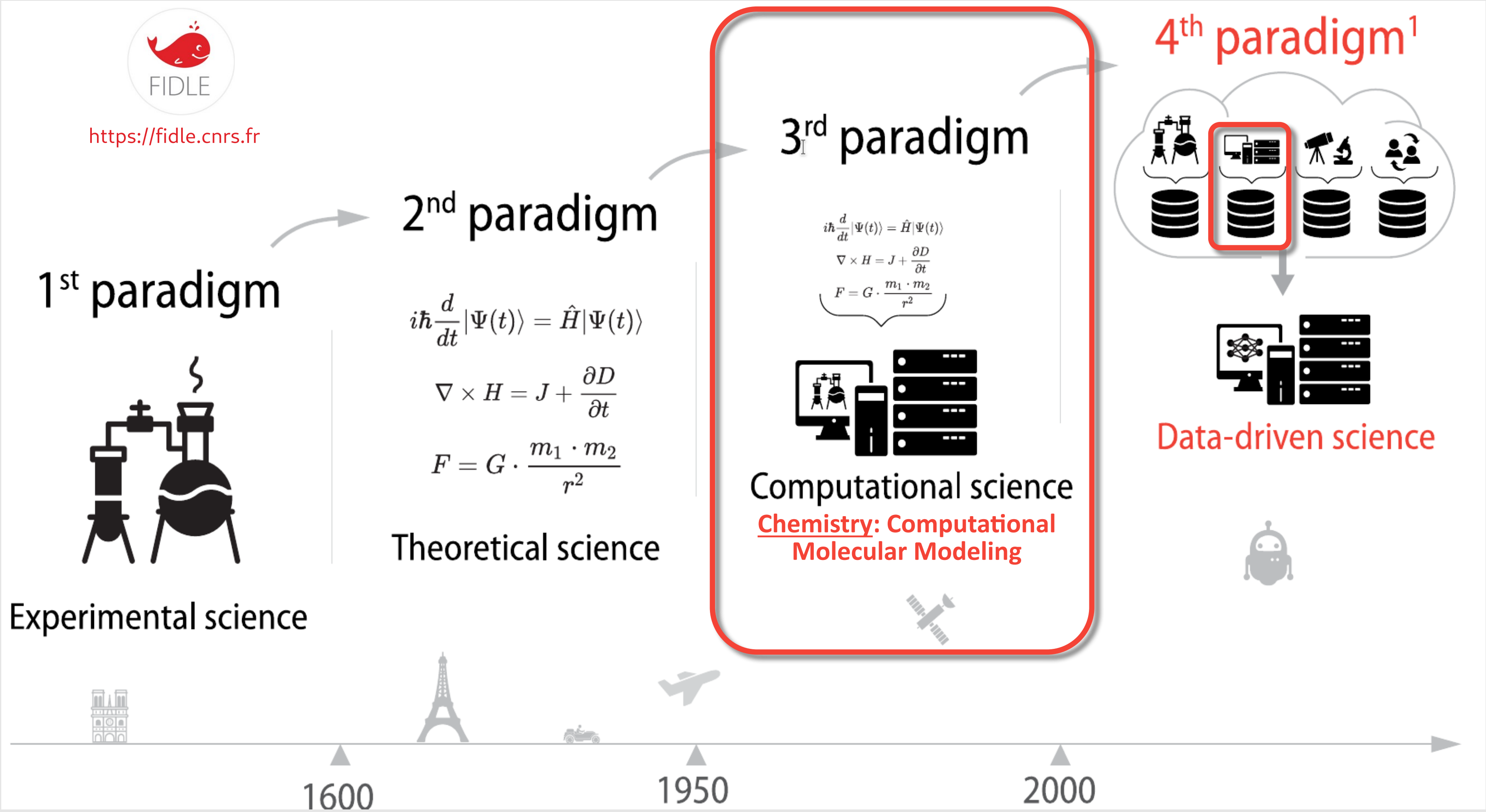
several single molecule properties account for those observed at the macroscopic level (spectroscopy, internal energy *U*(0K),...)

complements the information obtained by chemical experiments

in some cases predict hitherto unobserved chemical phenomena

impossible to describe a collection in molecules with all the complexity of their environment
→ simple molecular models and approximate methods are required

The four paradigms of scientific research and discovery



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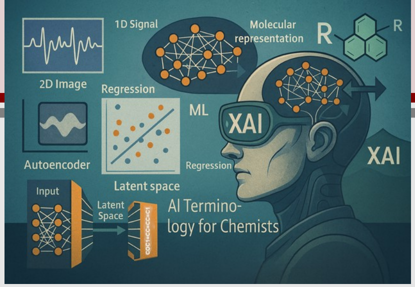
Machine Learning in brief



What did the model learn?

Which data ?

How much?

Collect data in a convenient form

Explainable AI tools to see through the *black box* models

Validate model performance on new data

Analyze and format data

Hidden patterns?

Train and optimize models

Which algorithm?

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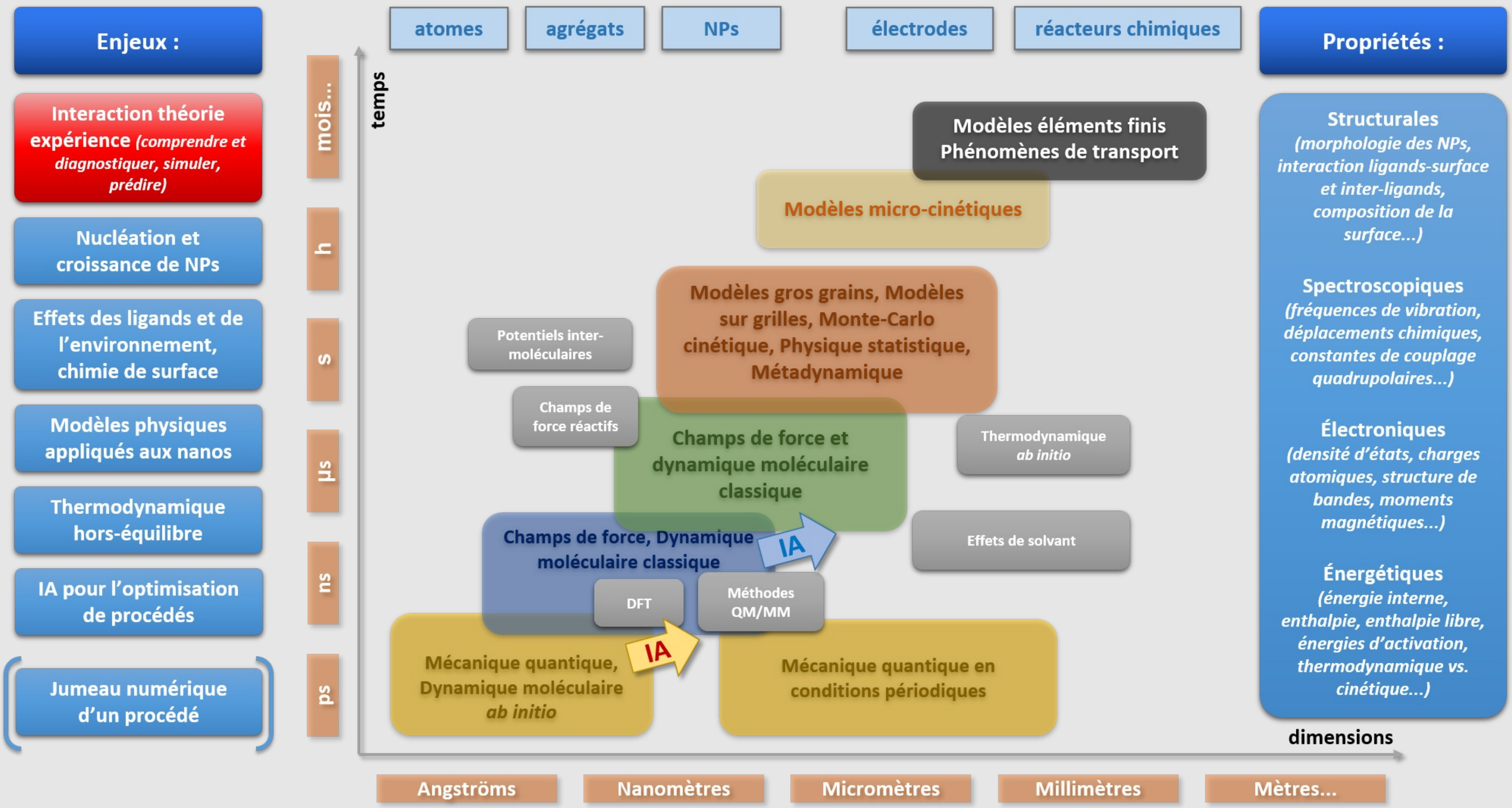
Il existe diverses techniques de modélisation dont le choix dépend de l'échelle temporelle et de la dimension spatiale des systèmes et processus étudiés



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Interest of virtual 3D models w.r.t. 3D physical models

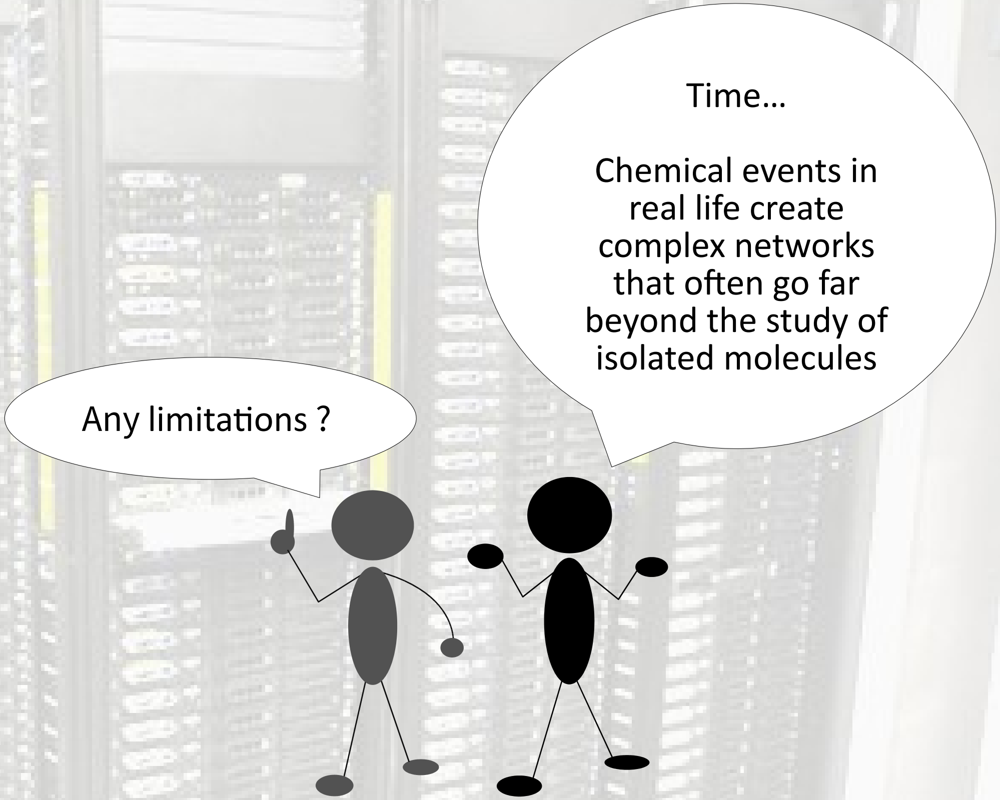
For a given geometry

***E* ⇒ Thermodynamics & Kinetics (TS Theory)**
***U*(0K)**
***H*°(*T*)**
***G*°(*T*)**
HOMO/LUMO

***Ψ* or *n* ⇒ Chemical bonding and interactions**
BDE
Nature of chemical bonds
Charge transfer
Dipole moment

***Ψ* or *n* ⇒ Spectroscopic properties**
X-ray
IR/Raman
NMR (quantum chemistry only)
UV-Vis (quantum chemistry only)

Solvent effects
Reaction pathways
Molecular Dynamics
…



Compromise computational cost / accuracy is necessary

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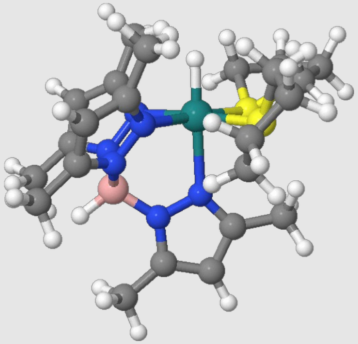
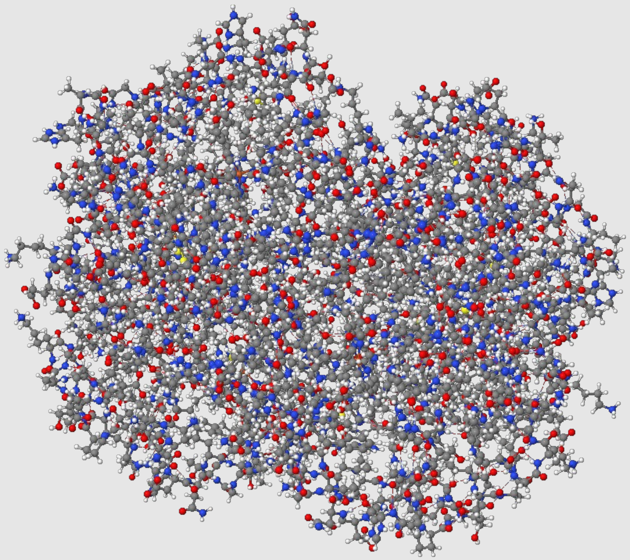
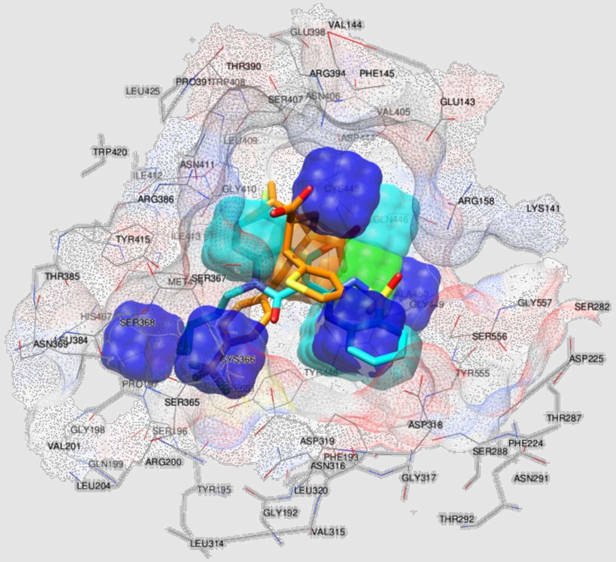
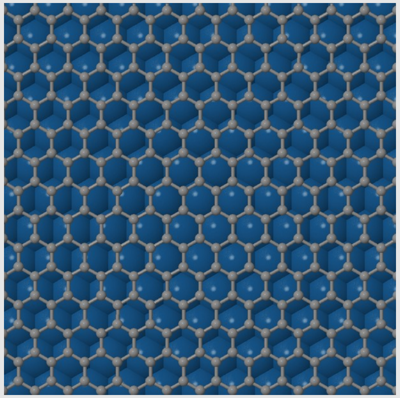
Which modelling method?

	Computational Quantum Mechanics (DFT, WFT, empirical, semi-empirical)	Empirical Force Field (FF) Models	Structure-Activity Relationships, Big Data, Artificial Intelligence (AI) & Machine learning
Molecular systems	 ruthenium complex	 human hemoglobin	 docking
Periodic systems	 graphene*Iridium		

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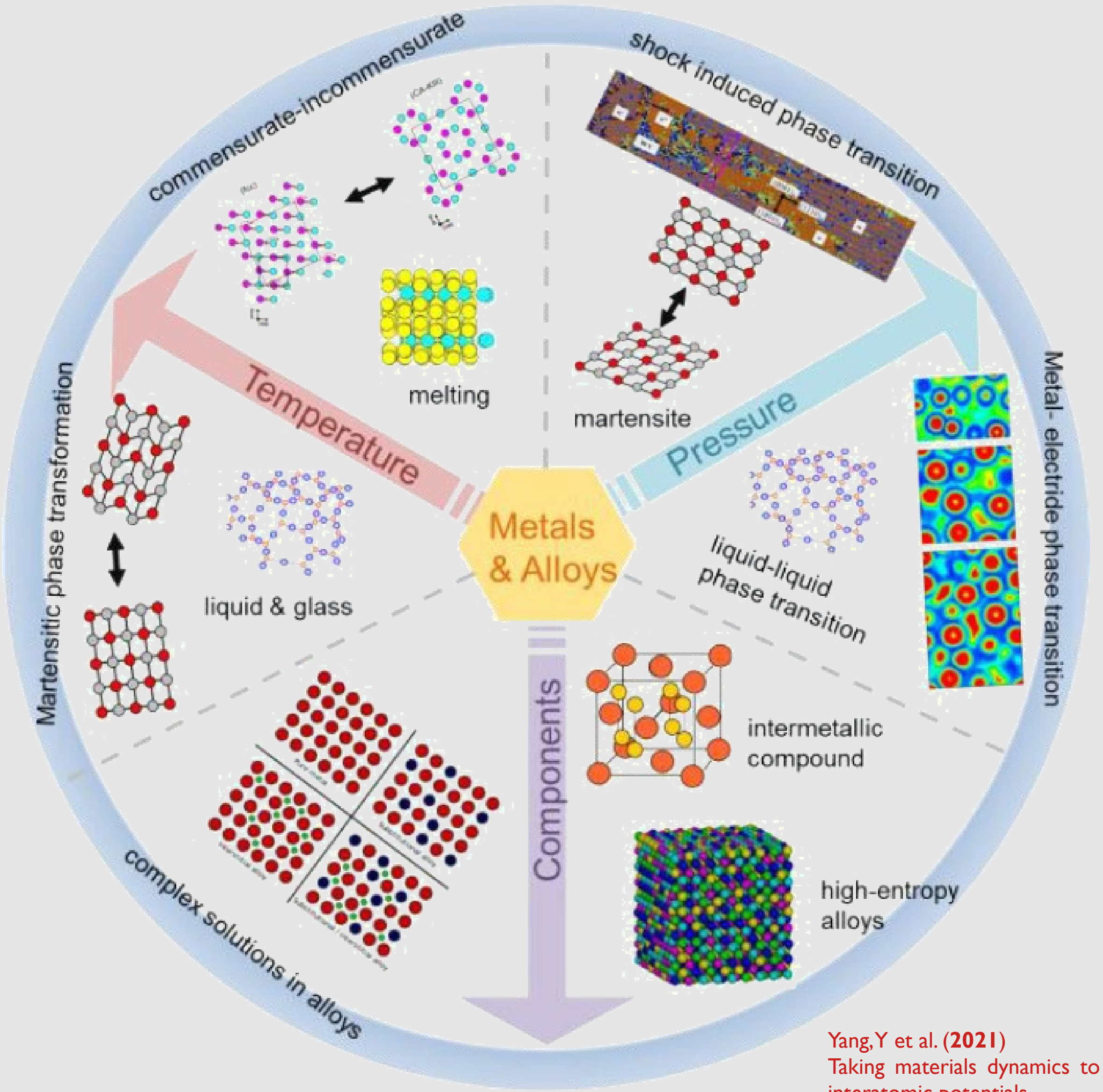
Machine Learning Interaction Potentials (MLIP)



Yang, Y et al. (**2021**)
Taking materials dynamics to new extremes using machine learning interatomic potentials.
J. Mater. Inf. **1**:10, 10.20517/jmi.2021.001

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Which modelling method?

Computational Quantum Mechanics

3D structure, isomerism
bond energies
chemical bond analysis
reactivity (*i.e.* bond breaking)
spectroscopy
docking
solvation
~~protein folding~~

10^2-10^3 atoms

Empirical Force Field Models & Machine Learning Interaction Potentials (MLIP)

3D structure, isomerism
~~bond energies~~
~~chemical bond analysis~~
~~reactivity (*i.e.* bond breaking)~~
spectroscopy (vibrational)
docking
solvation
protein folding

10^4-10^5 atoms

Structure-Activity Relationships, Big Data, Artificial Intelligence (AI) & Machine learning

Information retrieval
Drug design, Pharmacophores

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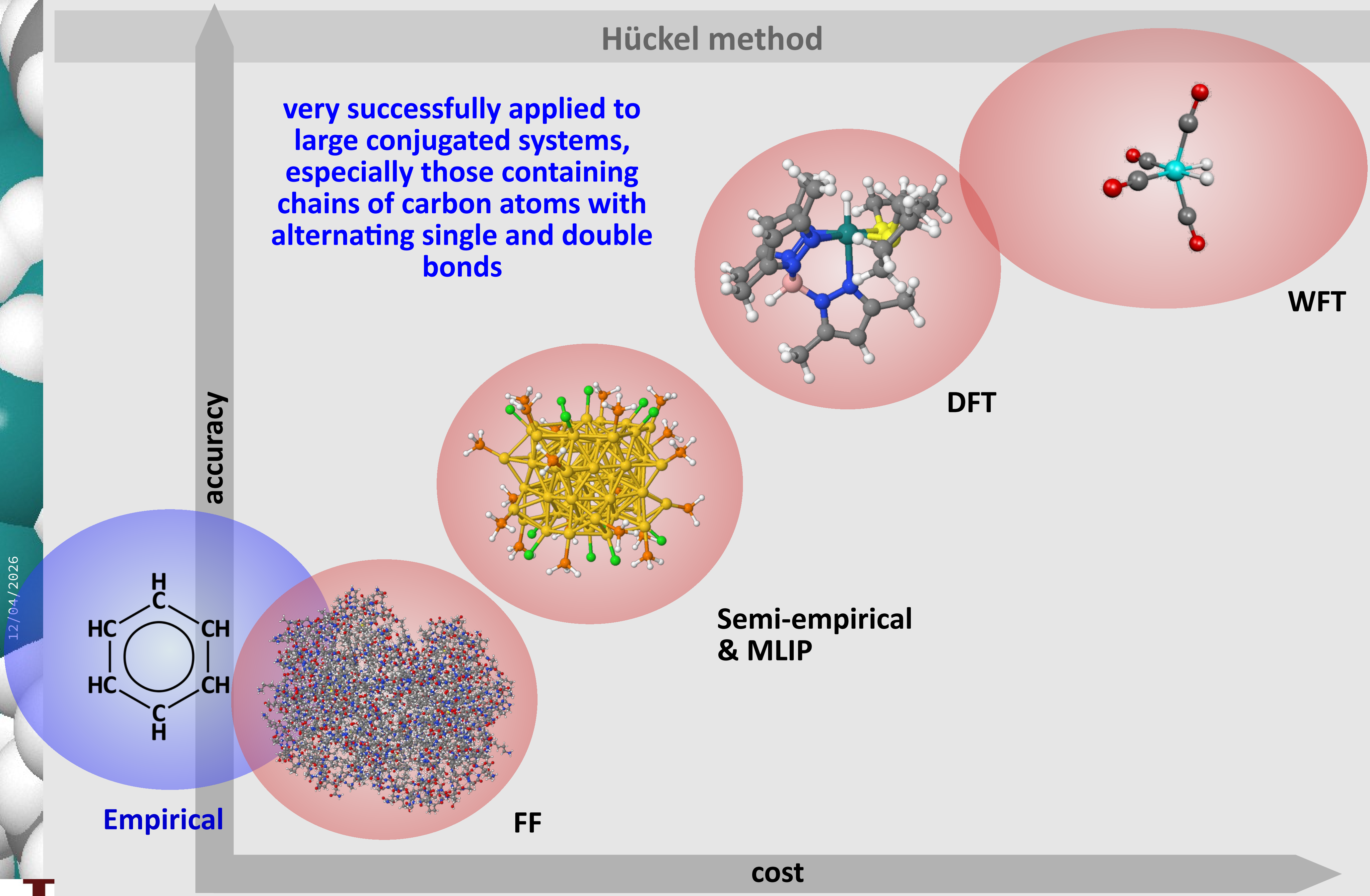
Expected accuracy / computational cost



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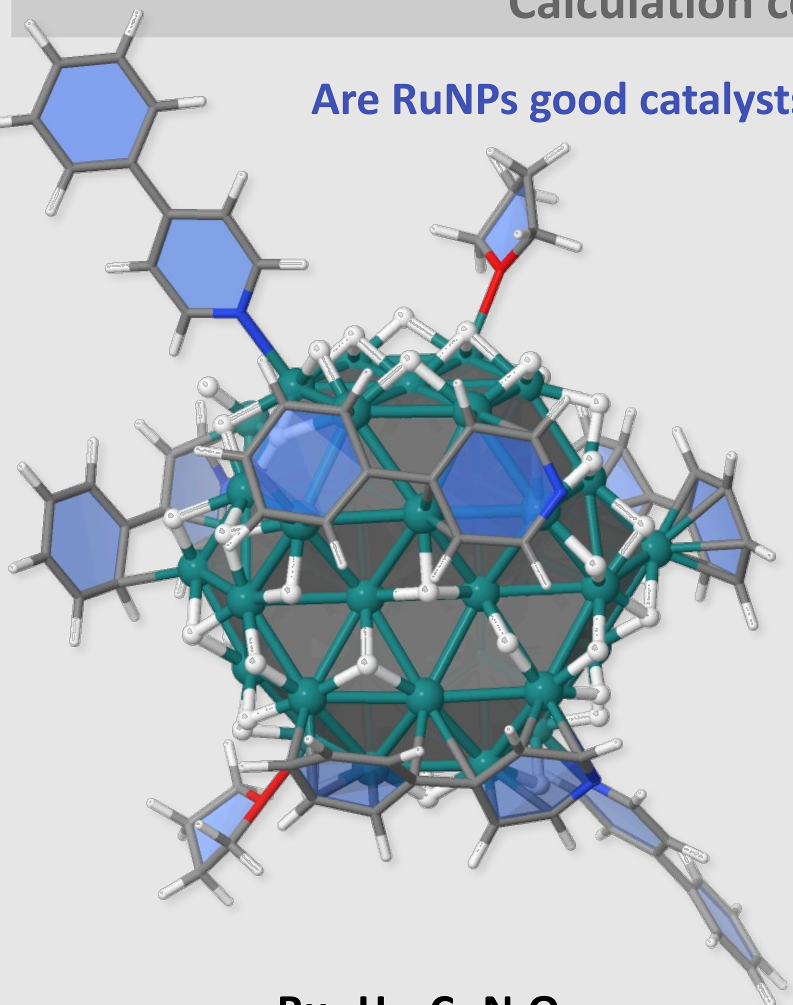
Calculation cost for a large molecular system

Are RuNPs good catalysts for the Hydrogen Evolution Reaction? $H^+ + e^- \rightarrow \frac{1}{2} H_2$



$Ru_{57}H_{123}C_{74}N_6O_2$

~ 10 geometry optimizations in order to calculate $\Delta_{ads}G(H)$

- took 200 000h (CPU time)

- each optimization runs on 144 cores (parallel calculation)

- 3 simultaneous calculations (432 cores)

⇒ ~ 500h real time = 20 days

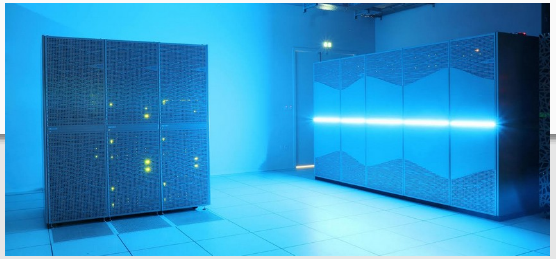


Photo credit: Dieter Gorecki

the computation cost is a serious bottleneck

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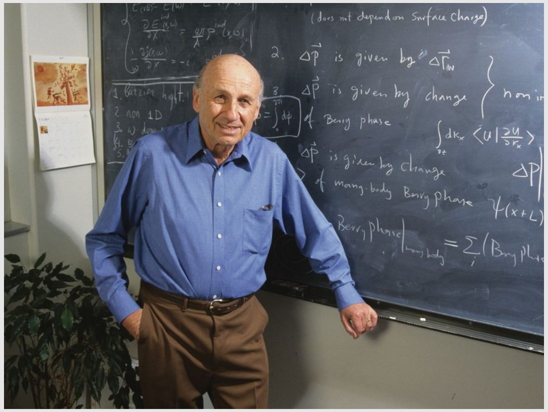
DFT: fundamental aspects



Walter Kohn (1923–2016)

– **DFT aims at solving the Schrödinger equation**
  $\hat{H}\Psi(\mathbf{r};\mathbf{R}) = E(\mathbf{R})\Psi(\mathbf{r};\mathbf{R})$
  of atoms, molecules, liquids & solids (condensed phase) in their <u>ground electronic state</u>

  where: $\hat{H} = \text{Kinetic Energy Operator} + \text{Potential Energy Operator}$

– **DFT is among the most popular and versatile methods available in condensed-matter physics, computational physics, and computational chemistry**

– **The energy E_0 of the ground state is a functional of the electron density n_0(r) that describes the electronic ground state**

$$E_0 = E[n_0(\mathrm{r})]$$

– **n_0(r) tells us everything we need to know about the system (energy, orbitals, dipole moment, polarisability, and all related spectroscopic properties)**

– **Theorem: the ground state wavefunction Ψ_0 is a unique functional of n_0:**

$$\Psi_0 = \Psi[n_0(\mathrm{r})]$$

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DFT: fundamental aspects

- **Problem: the functional that connects E_0 and $n_0(\mathrm{r})$ is unknown**

- **Approximations exist which permit the calculation of *E* and certain physical quantities quite accurately**

- **Roughly, several <u>approximations</u> are proposed by quantum chemists to solve DFT equations; each approximation provides a new <u>DFT functional</u>**

several approximate functionals are available in QC softwares:

PBE, TPSS, B3LYP, PBE0, B97, HSE06, ...

"the alphabet soup of approximate functionals"

K. Burke (2012) Perspective on density functional theory
J. Chem. Phys. **136**, 150901

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DFT: fundamental aspects

Successes of DFT

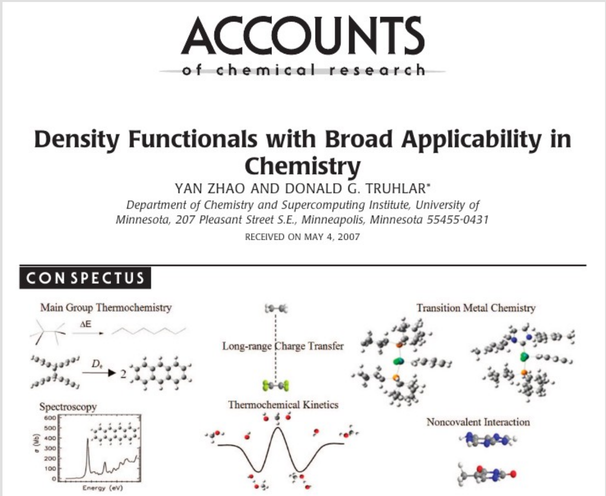
Molecular structures often in very close agreement with exp. (X-Ray)

Accurate reaction energetics (kin. & thermo.)

Vibrational frequencies also agree well with IR or Raman exp. spectra

^{1}H or ^{13}C NMR chemical shifts generally in agreement with exp. data

Vertical electronic transition energies often in good agreement with VUV exp. spectra


ACCOUNTS
of chemical research

Density Functionals with Broad Applicability in Chemistry

YAN ZHAO AND DONALD G. TRUHLAR*

Department of Chemistry and Supercomputing Institute, University of Minnesota, 207 Pleasant Street S.E., Minneapolis, Minnesota 55455-0431

RECEIVED ON MAY 4, 2007

CONSPECTUS

Failures of DFT

Does not account for dispersion forces (van der Waals)

Wrong description of charge-transfer excited states, or Rydberg excited states

Ill-behaviour if strongly correlated electrons (magnets etc...)

electron density for 2 uncorrelated electrons

$$n[1,2] \sim n[1]n[2]$$

electron density for 2 correlated electrons

$$n[1,2] \neq n[1]n[2]$$

Yet, DFT is based on the first approximation... :-(

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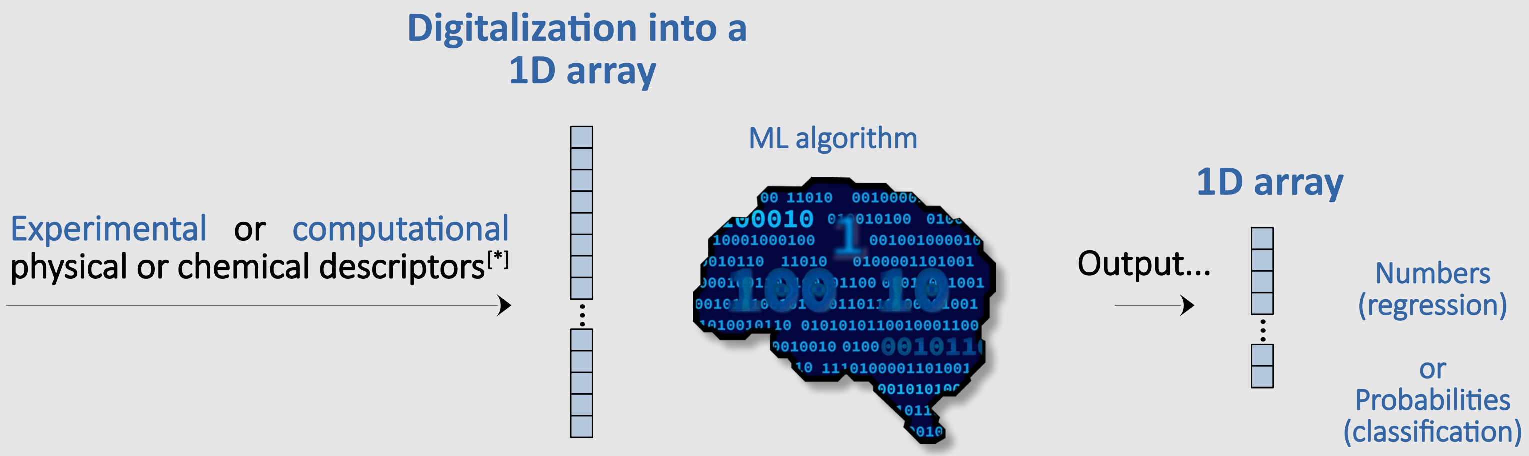
Zhao and TRuhlar (2007), *Acc. Chem. Res.* **41**, 157

Machine learning in physical chemistry (including or not DFT data)



[*] electron-counting rules, local aromaticity, HOMO-LUMO gaps, d-band centers, Fermi energy, global or local structural parameters, atomic charges, electron density, BDE, experimental spectrum, kinetic curve, thermal conductivity, heat capacity, resistivity, ductility,....

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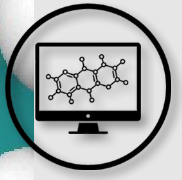
Conclusion

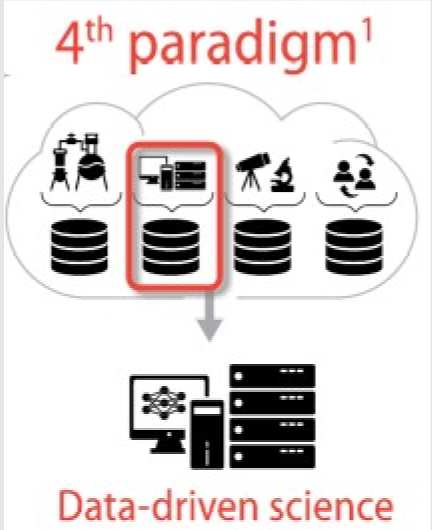


Necessity to maintain a DFT core expertise in standard application-driven "holistic" DFT to ensure a balanced and robust theoretical framework

Data Factory & High-Throughput: Implementing high-throughput DFT pipelines to transform theory into massive, AI-ready datasets

Electronic Descriptors: Moving beyond simple geometric criteria by injecting the richness of quantum electronic properties into the core of ML algorithms

Paradigm Shift: Transitioning DFT from a "direct calculation tool" to a "verification tool and learning engine" for predictive models



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